

Practical Network Traffic Analysis Using Wireshark

Student Name: Hashem Alarbeed

University: University of the people

Course: Advanced Networking and Data Security

Instructor: Dr. Zaharaddeen Sani

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a. TCP 3-way handshake interaction

No.	Time	Source	Destination	Protocol	Length	Info
5889	137.913995100	10.214.242.222	172.66.147.243	TCP	54	54733 → 80 [FIN, ACK] Seq=1 Ack=2 Win=131584 Len=0
5890	137.926816500	150.171.110.53	10.214.242.222	TCP	54	443 → 54737 [ACK] Seq=7176 Ack=2918 Win=77312 Len=0
5891	137.931506000	150.171.28.11	10.214.242.222	TLSv1.2	92	Application Data
5892	137.935388000	172.66.147.243	10.214.242.222	TCP	54	80 → 54734 [RST] Seq=2 Win=0 Len=0
5893	137.935388000	172.66.147.243	10.214.242.222	TCP	54	80 → 54733 [RST] Seq=2 Win=0 Len=0
5897	137.977891600	10.214.242.222	150.171.28.11	TCP	54	54738 → 443 [ACK] Seq=3898 Ack=262 Win=132096 Len=0
5899	140.358757000	10.214.242.222	172.66.147.243	TCP	66	54739 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
5900	140.363996500	10.214.242.222	172.66.147.243	TCP	66	54740 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
5901	140.386596200	172.66.147.243	10.214.242.222	TCP	66	80 → 54739 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SACK_PERM WS=8192
5902	140.386596200	172.66.147.243	10.214.242.222	TCP	66	80 → 54740 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SACK_PERM WS=8192
5903	140.386598000	10.214.242.222	172.66.147.243	TCP	54	54739 → 80 [ACK] Seq=1 Ack=1 Win=131584 Len=0
5904	140.386725900	10.214.242.222	172.66.147.243	TCP	54	54740 → 80 [ACK] Seq=1 Ack=1 Win=131584 Len=0
5905	140.387043900	10.214.242.222	172.66.147.243	HTTP	498	GET / HTTP/1.1
5906	140.404923700	172.66.147.243	10.214.242.222	TCP	54	80 → 54740 [ACK] Seq=1 Ack=445 Win=131072 Len=0
5907	140.413094300	172.66.147.243	10.214.242.222	TCP	736	80 → 54740 [PSH, ACK] Seq=1 Ack=445 Win=131072 Len=682 [TCP PDU reassembled in 5908]
5908	140.413994300	172.66.147.243	10.214.242.222	HTTP	59	HTTP/1.1 200 OK (text/html)
5909	140.413737000	10.214.242.222	172.66.147.243	TCP	54	54740 → 80 [ACK] Seq=445 Ack=688 Win=130816 Len=0
5912	140.505263500	10.214.242.222	172.66.147.243	HTTP	438	GET /favicon.ico HTTP/1.1
5913	140.531644200	172.66.147.243	10.214.242.222	TCP	678	80 → 54740 [PSH, ACK] Seq=688 Ack=829 Win=131072 Len=625 [TCP PDU reassembled in 5914]
5914	140.531644200	172.66.147.243	10.214.242.222	HTTP	59	HTTP/1.1 404 Not Found (text/html)
5915	140.531621500	10.214.242.222	172.66.147.243	TCP	54	54740 → 80 [ACK] Seq=829 Ack=1318 Win=130848 Len=0
5916	152.800728900	150.171.28.11	10.214.242.222	TLSv1.2	92	[TCP Previous segment not captured], Application Data
5917	152.800728900	150.171.28.11	10.214.242.222	TCP	743	[TCP Retransmission] 443 → 54738 [PSH, ACK] Seq=262 Ack=3898 Win=4194816 Len=689
5918	152.800728900	150.171.28.11	10.214.242.222	TCP	781	[TCP Retransmission] 443 → 54738 [PSH, ACK] Seq=262 Ack=3898 Win=4194816 Len=727
5919	152.800728900	150.171.28.11	10.214.242.222	TCP	781	[TCP Retransmission] 443 → 54738 [PSH, ACK] Seq=262 Ack=3898 Win=4194816 Len=727
5920	152.800728900	150.171.28.11	10.214.242.222	TCP	66	[TCP Dup ACK 5921#1] 54738 → 443 [ACK] Seq=3898 Ack=262 Win=132096 Len=0 SLE=951 SRE=909
5921	152.801086200	10.214.242.222	150.171.28.11	TCP	54	54738 → 443 [ACK] Seq=3898 Ack=989 Win=131328 Len=0
5922	152.801195000	10.214.242.222	150.171.28.11	TCP	66	[TCP Dup ACK 5921#1] 54738 → 443 [ACK] Seq=3898 Ack=989 Win=131328 Len=0 SLE=262 SRE=909
5923	152.801275200	10.214.242.222	150.171.28.11	TCP	66	[TCP Dup ACK 5921#2] 54738 → 443 [ACK] Seq=3898 Ack=989 Win=131328 Len=0 SLE=262 SRE=909
5924	152.805304900	10.214.242.222	150.171.28.11	TLSv1.2	96	Application Data
5925	152.824809500	150.171.28.11	10.214.242.222	TCP	54	443 → 54738 [ACK] Seq=989 Ack=3940 Win=4194816 Len=0
5928	154.231328300	10.214.242.222	74.125.29.139	TCP	66	54741 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
5929	154.267853900	74.125.29.139	10.214.242.222	TCP	66	443 → 54741 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
5930	154.267933900	10.214.242.222	74.125.29.139	TCP	54	54741 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=0
5931	154.268789900	10.214.242.222	74.125.29.139	TCP	1466	54741 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=1412 [TCP PDU reassembled in 5932]

Frame 5912: Packet, 438 bytes on wire (3504 bits), 438 bytes captured (3504 bits) on interface Device\NPF_{2A99E30A-1C2B-4096-B638-FE3BA3D492C}, id 0000 66 00 c7 26 29 c7 e4 42 a6 ea ef 39 08 00 45 00
Ethernet II, Src: Intel_ea:ef:39 (e4:42:a6:ea:ef:39), Dst: 66:00:c7:26:29:c7 (66:00:c7:26:29:c7)
Destination: 66:00:c7:26:29:c7 (66:00:c7:26:29:c7)
.....1..... : LG bit: Locally administered address (this is NOT the factory default)
.....0..... : LG bit: Individual address (unicast)

Explanation:

From my traffic analysis, I clearly noticed that the connection starts with three packets (SYN, SYN-ACK, ACK), and this was very clear around packet number 5899. Basically, the client sends a SYN packet first, then the server replies with SYN-ACK, and finally the client sends an ACK to complete the connection. This matches the explanation of how TCP establishes a reliable connection (Kurose & Ross, 2021).

b. MAC address of the PC

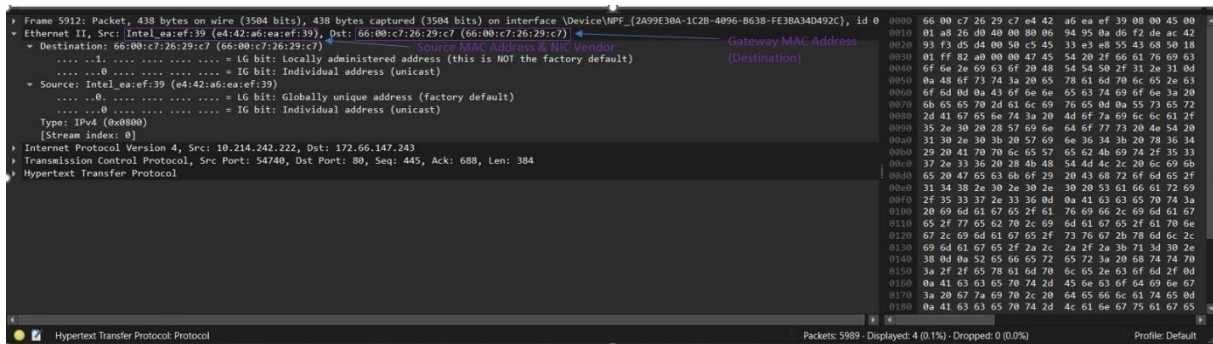
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5900	140.363996500	10.214.242.222	172.66.147.243	TCP	66	54740 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
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5903	140.386598000	10.214.242.222	172.66.147.243	TCP	54	54739 → 80 [ACK] Seq=1 Ack=1 Win=131584 Len=0
5904	140.386725900	10.214.242.222	172.66.147.243	TCP	54	54740 → 80 [ACK] Seq=1 Ack=1 Win=131584 Len=0
5905	140.387043900	10.214.242.222	172.66.147.243	HTTP	498	GET / HTTP/1.1
5906	140.404923700	172.66.147.243	10.214.242.222	TCP	54	80 → 54740 [ACK] Seq=1 Ack=445 Win=131072 Len=0
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5913	140.531644200	172.66.147.243	10.214.242.222	TCP	678	80 → 54740 [PSH, ACK] Seq=688 Ack=829 Win=131072 Len=625 [TCP PDU reassembled in 5914]
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5924	152.805304900	10.214.242.222	150.171.28.11	TLSv1.2	96	Application Data
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5928	154.231328300	10.214.242.222	74.125.29.139	TCP	66	54741 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
5929	154.267853900	74.125.29.139	10.214.242.222	TCP	66	443 → 54741 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
5930	154.267933900	10.214.242.222	74.125.29.139	TCP	54	54741 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=0
5931	154.268789900	10.214.242.222	74.125.29.139	TCP	1466	54741 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=1412 [TCP PDU reassembled in 5932]

Frame 5905: Packet, 498 bytes on wire (3984 bits), 498 bytes captured (3984 bits) on interface Device\NPF_{2A99E30A-1C2B-4096-B638-FE3BA3D492C}, id 0 0000 66 00 c7 26 29 c7 e4 42 a6 ea ef 39 08 00 45 00
Ethernet II, Src: Intel_ea:ef:39 (e4:42:a6:ea:ef:39), Dst: 66:00:c7:26:29:c7 (66:00:c7:26:29:c7)
Destination: 66:00:c7:26:29:c7 (66:00:c7:26:29:c7)
Source: Intel_ea:ef:39 (e4:42:a6:ea:ef:39) ← My MAC Address
Type: IPv4 (0x0800)
[Stream index 0]
Internet Protocol Version 4, Src: 10.214.242.222, Dst: 172.66.147.243
Transmission Control Protocol, Src Port: 54740, Dst Port: 80, Seq: 1, Ack: 1, Len: 444
Hypertext Transfer Protocol

Explanation :

While examining the Ethernet frame, I was able to identify my device's MAC address, and I noticed it appears as the source in all outgoing frames. This makes sense because the MAC address is a unique 48-bit identifier assigned to the network interface card (NIC) and operates at the data link layer (Stallings, 2016).

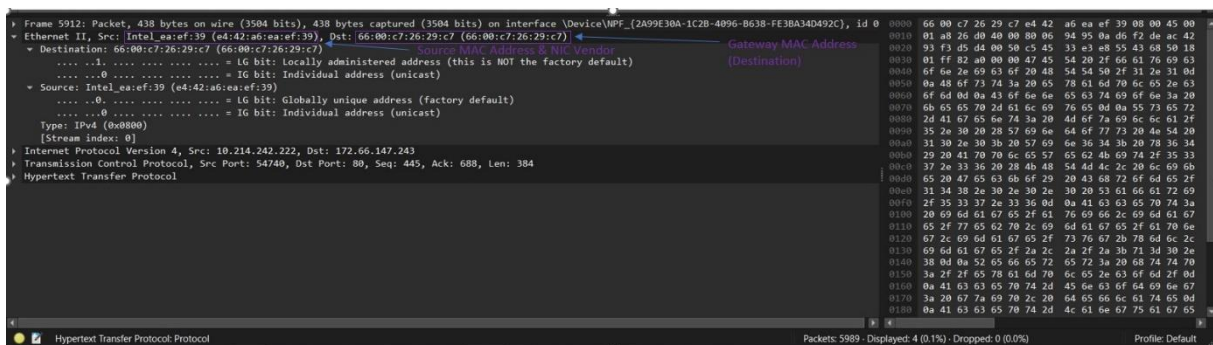
c. Network card vendor/manufacturer



Explanation:

By checking the OUI in Wireshark, I found that the manufacturer of my network card is Intel. From my understanding, the first 24 bits of the MAC address identify the vendor, which ensures global uniqueness of addresses (IEEE, 2023).

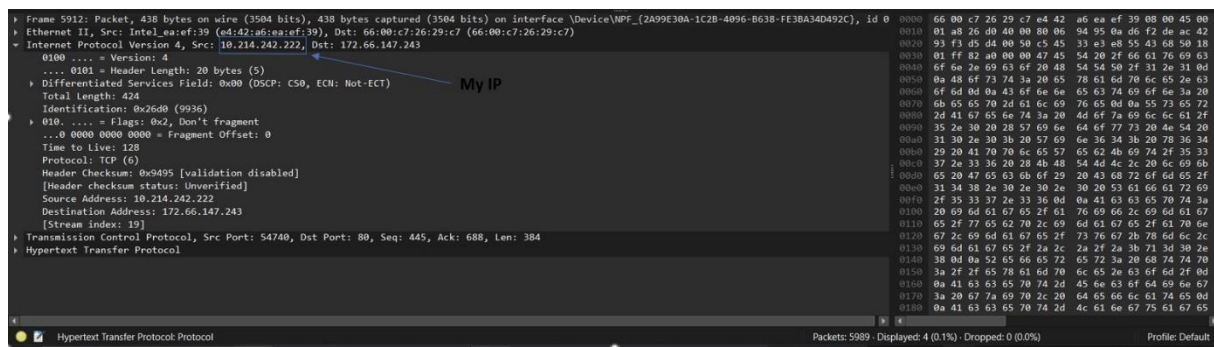
d. MAC address of the gateway device



Explanation:

Since I was accessing a website outside my local network, I noticed that the destination MAC address in outgoing frames belongs to the gateway. This is expected because the gateway forwards traffic from the local network to external networks (Lammle, 2020).

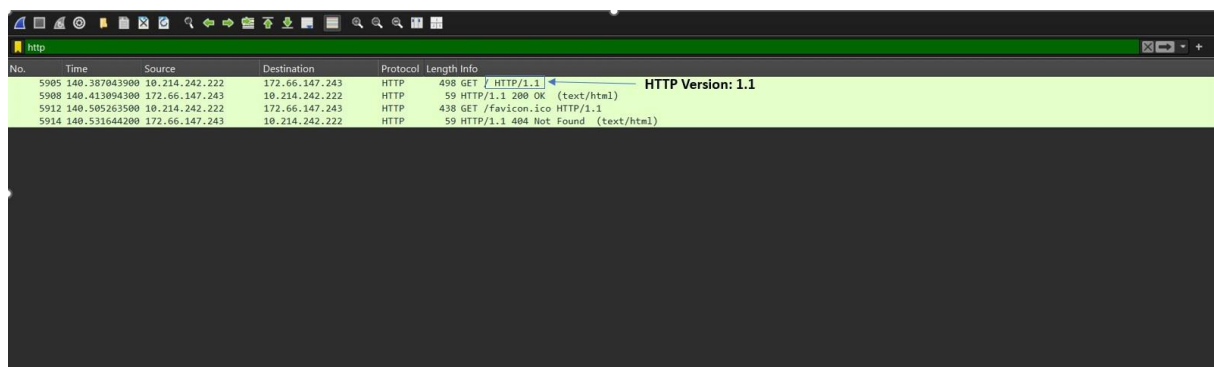
e. The IP address of the PC as shown in Wireshark



Explanation:

I identified my IP address as (10.214.242.222) from the captured packets. From what I understand, the IP address identifies the network interface rather than just the device itself, which aligns with standard networking concepts (Tanenbaum & Wetherall, 2021).

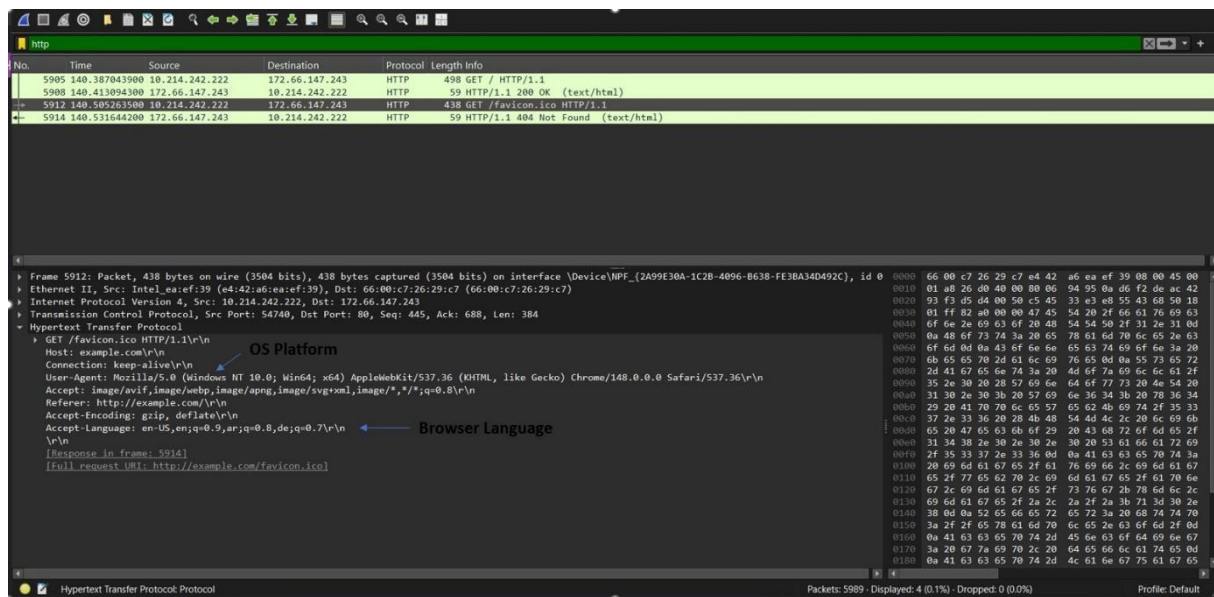
f. Version of HTTP captured



Explanation:

From the captured traffic, I observed that the browser is using HTTP/1.1. This version is still widely used for web communication and supports efficient data transfer between clients and servers (Fielding et al., 1999).

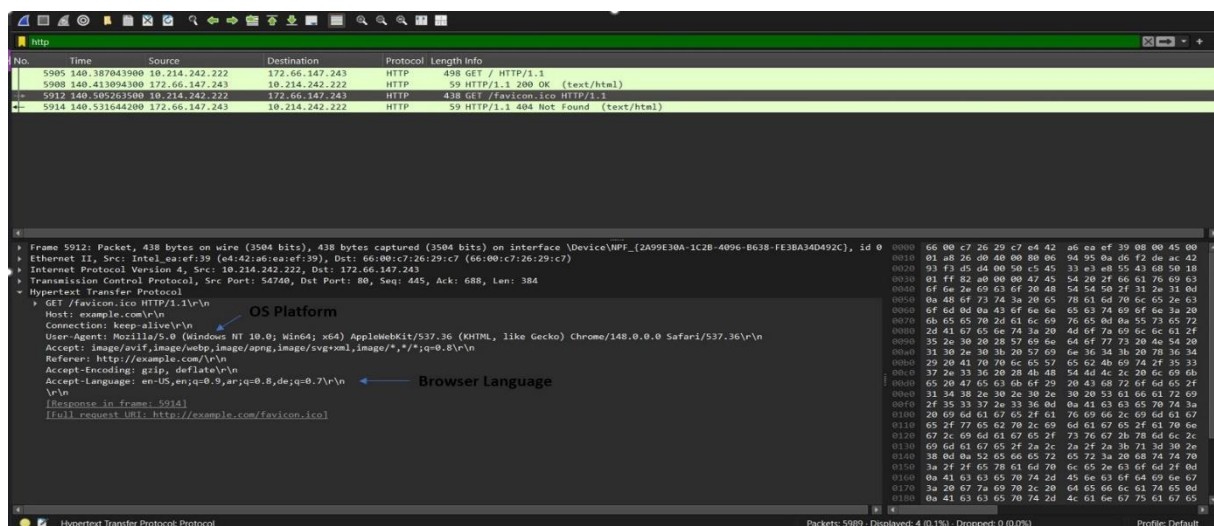
g. PC operating system (OS) platform



Explanation:

By analyzing the User-Agent field, I concluded that the operating system is Windows 10. The User-Agent string provides information about the client's platform and helps servers customize responses (MDN Web Docs, 2024).

h. PC browser language configuration



Explanation:

Finally, I found that the browser language is English (en-US), which was visible in the Accept-Language field. This field allows the browser to indicate preferred languages for content delivery (Reschke & Fielding, 2014).

Reference:

- Fielding, R., Gettys, J., Mogul, J., Frystyk, H., Masinter, L., Leach, P., & Berners-Lee, T. (1999). *Hypertext Transfer Protocol -- HTTP/1.1* (RFC 2616). Internet Engineering Task Force. <https://www.rfc-editor.org/rfc/rfc2616>
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